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In [5]: import pandas as pd
df=pd.read_csv('breast_cancer_clinical3.csv')
table1=pd.crosstab(df['ER'],df['Lymph_Node'])
table2=pd.crosstab(df['PR'],df['Lymph_Node'])
table3=pd.crosstab(df['HER2'],df['Lymph_Node'])
print('ER与LN的交叉表: ')
print(table1)
print('-----')
print('PR与LN的交叉表: ')
print(table2)
print('-----')
print('HER2与LN的交叉表')
print(table3)
print('-----')
import scipy
from scipy.stats import chi2_contingency
chi2,p,dof,expected=chi2_contingency(table1)
print('ER的chi2:',chi2)
print('ER的p值: ',p)
print('ER的dof: ',dof)
print('ER的expected:',expected)
print('-----')
chi2,p,dof,expected=chi2_contingency(table2)
print('PR的chi2:',chi2)
print('PR的p值:',p)
print('PR的dof: ',dof)
print('PR的expected:',expected)
print('-----')
chi2,p,dof,expected=chi2_contingency(table3)
print('HER2的chi2:',chi2)
print('HER2的p值:',p)
print('HER2的dof: ',dof)
print('HER2的expected:',expected)
print('-----')
print('chi square table:')
table4=pd.DataFrame({
    'variable':['ERvsLN','PRvsLN','HER2vsLN'],
    'chi2':[2.42,4.9,3.23],
    'p':[0.12,0.027,0.072]
})
print(table4)
print('PR的表达状态与淋巴结转移存在统计学关联')
print('ER和HER2的表达状态与淋巴结转移未发现显著关联')
table4.to_excel('chi2_analysis_table.xlsx')
print('-----')
from scipy.stats import fisher_exact
oddsratio,p=fisher_exact(table1)
print('ER与LN的OR:',oddsratio)
print('ER与LN的p值:',p)
print('-----')
oddsratio,p=fisher_exact(table2)
print('PR与LN的OR:',oddsratio)
print('PR与LN的p值:',p)
print('-----')
oddsratio,p=fisher_exact(table3)
print('HER2与LN的OR: ',oddsratio)
print('HER2与LN的p值:',p)
print('-----')

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table5=pd.DataFrame({
    'variable':['ERvsLN','PRvsLN','HER2vsLN'],
    'OR':[0.1,0.63,9.33],
    'p':[0.07,0.02,0.07]
})
print('fisher_exact table:')
print(table5)
print('PR的表达状态与淋巴结转移存在统计学关联')
print('ER与HER2的表达状态与淋巴结转移未发现明显关联')
table5.to_excel('fisher_exact_table.xlsx')
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ER与LN的交叉表:

Lymph_Node	0	1
ER		
0	1	6
1	8	5

PR与LN的交叉表:

Lymph_Node	0	1
PR		
0	2	9
1	7	2

HER2与LN的交叉表

Lymph_Node	0	1
HER2		
0	7	3
1	2	8

ER的chi2: 2.4175824175824174

ER的p值: 0.11997996734471256

ER的dof: 1

ER的expected: [[3.15 3.85]

[5.85 7.15]]

PR的chi2: 4.899500051015203

PR的p值: 0.02686447196134021

PR的dof: 1

PR的expected: [[4.95 6.05]

[4.05 4.95]]

HER2的chi2: 3.2323232323232327

HER2的p值: 0.07219819770165818

HER2的dof: 1

HER2的expected: [[4.5 5.5]

[4.5 5.5]]

chi square table:

	variable	chi2	p
0	ERvsLN	2.42	0.120
1	PRvsLN	4.90	0.027
2	HER2vsLN	3.23	0.072

PR的表达状态与淋巴结转移存在统计学关联

ER和HER2的表达状态与淋巴结转移未发现显著关联

ER与LN的OR: 0.10416666666666667

ER与LN的p值: 0.07027863777089784

PR与LN的OR: 0.06349206349206349

PR与LN的p值: 0.02155275065491784

HER2与LN的OR: 9.333333333333334

HER2与LN的p值: 0.06977851869492736

fisher_exact table:

	variable	OR	p
0	ERvsLN	0.10	0.07
1	PRvsLN	0.63	0.02
2	HER2vsLN	9.33	0.07

PR的表达状态与淋巴结转移存在统计学关联

ER与HER2的表达状态与淋巴结转移未发现明显关联

In []: